

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414871
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Gage; Shawn M. et al.

Apparatus and method for implanting an arteriovenous graft

Abstract

An apparatus is provided for subcutaneous implantation in a patient using a tunneling instrument. The implantation apparatus comprises a vascular graft and a connector adapted to couple a distal end of the tunneling instrument and a proximal end of the graft. The connector comprises a tip, a first end of the tip configured to be received within the proximal end of the graft, a clip for securing the graft to the tip, and a coupler for a rotatable connection of the tip to the tunneling instrument such that the tip is rotatable about its longitudinal axis relative to the coupler to facilitate attachment of the graft to the tunneling instrument. The implantation apparatus may further comprise a removable sheath configured to substantially cover the length of the graft.

Inventors: Gage; Shawn M. (Raleigh, NC), Knight; Joseph (Durham, NC), Lawson; Michael (Durham, NC), Nichols; Craig (Durham, NC)

Applicant: INNAVASC MEDICAL, INC. (Durham, NC)

Family ID: 1000008820798

Assignee: InnAVasc Medical, Inc. (Durham, NC)

Appl. No.: 17/360880

Filed: June 28, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210322193 A1	Oct. 21, 2021

Related U.S. Application Data

division parent-doc US 16723211 20191220 US 11045339 child-doc US 17360880
us-provisional-application US 62940438 20191126
us-provisional-application US 62783187 20181220

Publication Classification

Int. Cl.: A61F2/966 (20130101); A61F2/95 (20130101)

U.S. Cl.:

CPC A61F2/966 (20130101); A61F2/9522 (20200501); A61F2002/9665 (20130101)

Field of Classification Search

CPC: A61F (2/95); A61F (2/962); A61F (2/966); A61F (2002/9505); A61F (2002/9522); A61F (2002/9665)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5306240	12/1993	Berry	N/A	N/A
6254628	12/2000	Wallace et al.	N/A	N/A
7628795	12/2008	Karwoski	N/A	N/A
7727268	12/2009	Cunniffe et al.	N/A	N/A
8177832	12/2011	Armstrong et al.	N/A	N/A
8292938	12/2011	Case et al.	N/A	N/A
8545544	12/2012	Spenser et al.	N/A	N/A
8622886	12/2013	Young et al.	N/A	N/A
9125657	12/2014	Ray et al.	N/A	N/A
9345566	12/2015	Bosel	N/A	N/A
9387101	12/2015	Dorn et al.	N/A	N/A
9622893	12/2016	Huser	N/A	N/A
9687369	12/2016	Dorn et al.	N/A	N/A
9808366	12/2016	Dorn	N/A	N/A
10045868	12/2017	Cully et al.	N/A	N/A
10149750	12/2017	Wagner et al.	N/A	N/A
10271979	12/2018	Dorn et al.	N/A	N/A
10328243	12/2018	Spear et al.	N/A	N/A
11045339	12/2020	Gage et al.	N/A	N/A
2003/0125789	12/2002	Ross et al.	N/A	N/A
2005/0192604	12/2004	Carson et al.	N/A	N/A
2005/0203457	12/2004	Smego	N/A	N/A
2006/0015171	12/2005	Armstrong	N/A	N/A
2006/0135949	12/2005	Rome et al.	N/A	N/A
2006/0173467	12/2005	Karwoski et al.	N/A	N/A
2009/0192586	12/2008	Tabor et al.	N/A	N/A
2013/0304185	12/2012	Newell	N/A	N/A
2017/0072129	12/2016	Slager et al.	N/A	N/A
2018/0078353	12/2017	Alexander et al.	N/A	N/A
2018/0193178	12/2017	Zukowski et al.	N/A	N/A
2018/0264188	12/2017	Ash	N/A	N/A
2020/0197205	12/2019	Gage et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2016196362	12/2015	WO	N/A
2020132505	12/2019	WO	N/A

OTHER PUBLICATIONS

Gage, Shawn M.; Non-Final Office Action for U.S. Appl. No. 16/723,211, filed Dec. 20, 2019, mailed Mar. 10, 2020, 22 pages. cited by applicant

Gage, Shawn M.; Notice of Allowance for U.S. Appl. No. 16/723,211, filed Dec. 20, 2019, mailed Mar. 1, 2021, 9 pgs. cited by applicant

Gage, Shawn W.; Non-Final Office Action for U.S. Appl. No. 16/723,211, filed Dec. 20, 2019, mailed Oct. 19, 2020, 10 pgs. cited by applicant

Gage; Shawn M.; Notice of Allowance for U.S. Appl. No. 16/723,211, filed Dec. 20, 2019, mailed Jul. 1, 2020, 11 pages. cited by applicant

Gage, Shawn; International Search Report and Written Opinion for PCT Application No. PCT/US2019/067954, filed Dec. 20, 2019, mailed Jul. 2, 2020, 23 pages. cited by applicant

Gage, Shawn; International Search Report and Written Opinion for PCT/US21/65756, filed Dec. 30, 2021, mailed Apr. 7, 2022, 18 pgs. cited by applicant

Primary Examiner: Fishback; Ashley L

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/723,211, filed Dec. 20, 2019, which claims the benefit of U.S. Provisional Application No. 62/783,187, filed Dec. 20, 2018, and U.S. Provisional Application No. 62/940,438, filed Nov. 26, 2019, the contents of all of which are incorporated herein in their entirety.

BACKGROUND

(1) An apparatus and method are described for implanting an arteriovenous graft and, more particularly, an apparatus and method for use in forming a subcutaneous anatomical tunnel for implanting the arteriovenous graft, including use of a removable sheath enclosing the graft during implantation.

(2) An arteriovenous graft is a tubular device that is suitable for implantation in the body to redirect flow of blood between blood vessels. Surgical implantation of the arteriovenous graft requires placement of the graft within subcutaneous tissue. An initial step in the implantation procedure is the creation of a subcutaneous anatomic pathway for the arteriovenous graft, which is commonly called a graft tunnel, between anastomotic sites for passage of the vascular graft. This is a required surgical step in peripheral vascular procedures for all peripheral, vascular access and extra-anatomical graft locations. The arteriovenous graft is positioned in the tunnel within the bodily tissue for fixation of the graft to an existing peripheral vessel to form a bypass around the vessel, or a portion thereof, or connection of an artery and vein to form an arteriovenous shunt. The vascular graft may also connect an artery to an artery.

(3) A conventional tunneling device includes an elongated rigid rod having a handle on a proximal end and a bullet-shaped tip at a distal end. The rod may vary in size and shape and may have a straight shaft, a curved shaft or a semicircular shaft, which allows for a variety of graft placement positions and locations. In the tunneling procedure, a proximal incision and a distal incision are

made at a chosen area of anastomosis. The tip at the distal end of the tunneling device is inserted into the proximal incision. The tip of the tunneling device is then forcefully passed through the subcutaneous tissue creating a path between the incisions by blunt dissection until the tip protrudes from the distal incision. Once the tip is exposed, a proximal end of the arteriovenous graft is tied onto the distal end or the tip of the tunneling device with sterile suture thread. The tunneling instrument and attached arteriovenous graft are then pulled along the path through the recently dissected graft tunnel until the proximal end of the arteriovenous graft extends from the proximal incision. When the arteriovenous graft is appropriately positioned, the graft is cut free from the distal end of the tunneling instrument. An anastomosis is formed around the area of vasculature to be bypassed and the incisions are closed.

(4) The step of pulling the tunneling instrument and attached arteriovenous graft through the graft tunnel requires significant force. The force required depends on a number of factors, including the relative sizes of the graft tunnel and the graft and the material of the graft. Conventional delivery systems for venous and other implantable devices are sometimes covered by a retaining sheath that reduces the friction of passage through the subcutaneous tissue. Following implantation, the sheath is removed by rolling back over the device in order to retract the sheath. A pull member may be provided and connected to the sheath to retract the sheath. In one application, the sheath is folded back onto itself so as to provide an inner sheath and an outer sheath disposed over and extending axially along the device. The outer sheath is attached to the pull member. As the pull member is pulled, the outer sheath moves with it causing the sheath to “roll” with respect to the device, thereby progressively uncovering the device. However, while rolling a sheath during retraction reduces the necessary pulling force as compared to withdrawing the sheath by sliding the sheath over the device, there is still significant force necessary to retract a sheath following implantation of an arteriovenous graft.

(5) For the foregoing reasons, there is a need for an apparatus and method for implanting an arteriovenous graft for minimizing trauma to tissue. The arteriovenous graft may be enclosed within a flexible, expandable sheath configured to surround a length of the tunneling device or the graft. The sheath may be coated on an outside surface with a lubricious substance to provide a low coefficient of friction, allowing the sheath and tunneling device to be easily pushed through tissue. The sheath should be retractable in a reliable manner with a low pulling force for minimizing problems associated with excessive axial forces on the sheath during retraction. The tunneling device should be capable of use with any type of vascular graft including, but not limited to, a natural tissue graft. The apparatus may comprise a tunneling device that allows for delivery of fluid adjacent to the tip during the tunneling process.

SUMMARY

(6) An apparatus is provided for subcutaneous implantation in a patient using a tunneling instrument, including a shaft having a proximal end and a distal end. The implantation apparatus comprises a vascular graft having a proximal end and a distal end and a length between the proximal end and the distal end. A connector is adapted to couple the distal end of the tunneling instrument and the proximal end of the graft. The connector comprises a tip, a first end of the tip configured to be received within the proximal end of the graft, a clip for securing the graft to the tip, and a coupler for a rotatable connection of the tip to the tunneling instrument such that the tip is rotatable about its longitudinal axis relative to the coupler to facilitate attachment of the graft to the tunneling instrument.

(7) In one aspect, the coupler comprises a screw thread formed on an external surface for connection of the coupler to the tunneling instrument. In a further aspect, the coupler comprises a ferrule defining an axial passage for rotatably receiving a proximal end of the tip.

(8) In one embodiment, the implantation apparatus further comprises a removable sheath configured to substantially cover the length of the graft.

(9) In another embodiment, the tip comprises a body including an intermediate portion having a

reduced diameter forming a circumferential groove around a periphery, and wherein the clip is positioned in the groove, a portion of the graft being positioned between the clip and the tip within the groove, the clip being crimped about the tip to fix the graft to the tip. The clip is sized and shaped to provide a snap-fit connection within the groove.

(10) In an alternative embodiment, the tip is adapted to be removably attached to the distal end of the tunneling instrument, and wherein the connector is configured for connecting the graft to the tunneling instrument upon removal of the tip. In one aspect, the coupler comprises a frustoconical cone having an axial opening for receiving the graft, a length of the cone including a screw thread formed on an external surface for connection of the cone to the tunneling instrument, wherein a portion of the graft is positioned between the threads of the cone and the tunneling instrument to fix the graft to the tip. In another aspect, the coupler comprises a hollow sleeve for rotatable connection to the tunneling instrument, and a plug received within the proximal end of the graft. The sleeve is configured for receiving the plug such that a portion of the graft is positioned between the plug and the sleeve to fix the graft in the sleeve.

(11) The coupler is adapted to be attached to a synthetic vascular graft, a natural tissue vascular graft or an arteriovenous graft.

(12) An implantable device is also provided for subcutaneous delivery in a patient using a tunneling instrument including a shaft having a proximal end and a distal end. The implantable device comprises a vascular graft having a length, a distal end, a proximal end, an outer surface and a longitudinal axis. A sheath having a length is positioned over a substantial portion of the outer surface of the vascular graft. The sheath has longitudinally spaced perforations along the length of the sheath for permitting tearing of the sheath. The sheath is configured to evert upon application of a longitudinal force to the sheath following implantation of the vascular graft to cause the sheath to move in a proximal direction during removal of the sheath from the implanted graft. Upon eversion and exceeding a tearing strength, the sheath tears progressively along the perforations that are generally oriented in a linear arrangement along the length of the sheath.

(13) In one aspect, two rows of perforations are generally oriented circumferentially opposite to one another. In another aspect, two rows of perforations are generally oriented adjacent and parallel to one another.

(14) In one embodiment, the implantable device further comprises a tether connected to the sheath for applying a pulling force in a proximal direction. One end of the tether may be attached to the sheath at a position adjacent to the distal end of the sheath.

(15) In another embodiment, a sheath having perforations is folded back on itself forming a double wall having an inner portion and an outer portion. The sheath is configured to slide over itself upon application of a longitudinal force to the sheath following implantation of the vascular graft to cause the sheath to tear along the perforations while moving in a proximal direction during removal of the sheath from the implanted graft thus reducing the force necessary to extract the sheath.

(16) A tunneling device is provided for delivery of fluid from a fluid source for use in the subcutaneous placement of a vascular graft in a patient. The tunneling device comprises a rigid hollow rod for forming a subcutaneous path and having a distal end, a proximal end and an interior lumen defining a fluid pathway. The interior lumen is adapted to be in fluid communication with the fluid source. A hollow tapered tip is mounted to the distal end of the rod and has an interior lumen in fluid communication with the lumen in the rod. One of the rod or the tip defines at least one opening for delivery of fluid from the fluid source through the at least one opening.

(17) Another tunneling device is provided for use in the subcutaneous placement of a vascular graft for attachment to a blood vessel in a patient. The tunneling device comprises a rigid rod having a proximal end and a distal end for forming a subcutaneous path. A magnetic tapered tip is at the distal end of the rod. A magnetic wand magnetically engages the tip from external to the skin for longitudinal advancement of the rod and tip along the subcutaneous path.

(18) Yet another tunneling device is provided for use in the subcutaneous placement of a vascular

graft in a patient. The tunneling device comprises a rigid helical rod having a proximal end and a distal end. A tapered tip is mounted to the distal end of the rod. In use, the rod forms a helical tunnel when rotated during longitudinal advancement along a subcutaneous path.

(19) A method is provided for using a tunneling device having a distal tip for placement of a vascular graft along a subcutaneous path in a patient. The tunneling device method comprises the steps of forming an incision adjacent to a chosen area of anastomosis and placing markers on the skin indicating a predetermined subcutaneous path. The tunneling device is inserted in the incision for performing blunt dissection of subcutaneous tissue along the path indicated by the markers with the tunneling device. A proximal end of the graft is attached to the tunneling device followed by the step of retracting the tunneling device from the incision until the proximal end of the graft exits the incision. The graft is released from the tunneling device and anastomoses are formed of the ends of the graft to a first blood vessel and a second blood vessel and then the incision is closed. In one aspect, the tip of the tunneling device may be removed before attaching the graft.

(20) Still another tunneling device is provided for use in subcutaneous placement of a vascular graft in a patient. The tunneling device comprises a guide wire and a hollow inflatable conduit coaxially surrounding the guide wire over a substantial portion of its length. The conduit in a delivery configuration collapses against the wire due to confining pressure when inserted subcutaneously into a body. Inflation of the conduit radially expands the conduit to a deployed configuration and spaces the conduit from the wire over substantially the full length of the wire and preventing collapse of the subcutaneous tissue inwardly against the wire. The graft may be introduced into the conduit while the conduit is inflated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the apparatus and method for use in forming a subcutaneous anatomical tunnel, reference should now be had to the embodiments shown in the accompanying drawings and described below. In the drawings:
- (2) FIG. 1 is a side perspective view of a tunneling instrument including an embodiment of a tip assembly.
- (3) FIG. 2 is a close-up side perspective view of a distal portion of the tunneling instrument and the tip assembly as shown in FIG. 1.
- (4) FIG. 3 is a distal end perspective view of the portion of the tunneling instrument and the tip assembly as shown in FIG. 2.
- (5) FIG. 4 is a side elevation view of the tip assembly as shown in FIGS. 1-3.
- (6) FIG. 5 is an exploded side elevation view of the tip assembly as shown in FIG. 4.
- (7) FIG. 6 is a distal end perspective view of an embodiment of a tip for use with the tip assembly as shown in FIG. 4.
- (8) FIG. 7 is a side elevation view of the tip as shown in FIG. 6.
- (9) FIGS. 8 and 9 are distal and proximal end perspective views of an embodiment of a swivel connector for use with the tip assembly as shown in FIG. 4.
- (10) FIG. 10 is a front, side and rear perspective views of an embodiment of a graft clip for use in the tip assembly shown in FIG. 4.
- (11) FIG. 11 is a side elevation view of the tip assembly and graft clip as shown in FIG. 10.
- (12) FIG. 12 is an exploded side perspective view of the tip and graft clip as shown in FIG. 11.
- (13) FIG. 13 is a front perspective view of another embodiment of a graft clip in a first closed position for use with the tip assembly as shown in FIG. 4.
- (14) FIG. 14 is a front perspective view of the graft clip as shown in FIG. 13 with the graft clip in a second open position.

(15) FIG. 15 is a front perspective view of the graft clip as shown in FIG. 14 with the graft clip in the open position on the tip assembly as shown in FIG. 4.

(16) FIG. 16 is a front perspective view of the graft clip as shown in FIG. 13 with the graft clip in a second closed position on the tip assembly.

(17) FIG. 17 is a side perspective view of the graft clip as shown in FIG. 14 with the graft clip in the open position on the tip assembly as shown in FIG. 4 and surrounding a proximal end of an AVG.

(18) FIG. 18 is a side perspective view of the graft clip as shown in FIG. 13 with the graft clip in a second closed position on the tip assembly over the proximal end of the AVG.

(19) FIG. 19 is an exploded perspective view of an embodiment of a coupler for connecting a proximal end of an arteriovenous graft to a distal end of a tunneling device.

(20) FIG. 20 is a side perspective views of another embodiment of a tunneling device including a frustoconical hood showing steps for use as a coupler for connecting a proximal end of an arteriovenous graft to a distal end of a tunneling device.

(21) FIG. 21 is a perspective view of a sheath surrounding at least a substantial portion of an arteriovenous graft and secured with the AVG to the tip assembly and tunneling instrument as shown in FIG. 1 using the clip as shown in FIG. 13.

(22) FIG. 22 is a close-up of the perspective view of the sheath and arteriovenous graft connected to the tip assembly and tunneling instrument as shown in FIG. 21.

(23) FIGS. 23A-23C are schematic views of an arteriovenous graft enclosed within an embodiment of a removable sheath with a proximal end of the combined graft and sheath attached to the tip assembly on a tunneling instrument.

(24) FIGS. 24A and 24B are schematic views of an arteriovenous graft enclosed within another embodiment of a removable sheath with a proximal end of the combined graft and sheath attached to the tip assembly on a tunneling instrument.

(25) FIGS. 25A-25C are schematic views of an arteriovenous graft enclosed within a third embodiment of a removable sheath with a proximal end of the combined graft and sheath attached to the tip assembly on a tunneling instrument.

(26) FIG. 26 is a side perspective view of another embodiment of a tip assembly and tunneling instrument for use in forming a subcutaneous anatomical tunnel for implantation of a vascular graft.

(27) FIG. 27 is a side perspective view of a third embodiment of a tip assembly and tunneling instrument for use in forming a subcutaneous anatomical tunnel for implantation of a vascular graft.

(28) FIG. 28 is a side perspective view of a fourth embodiment of a tip assembly and tunneling instrument for use in forming a subcutaneous anatomical tunnel for implantation of a vascular graft including a magnetic tip portion and a wand.

(29) FIG. 29 is a schematic front elevation view of an embodiment of an apparatus for use in forming a subcutaneous anatomical tunnel for implantation of a vascular graft using a self-guided tunneling device.

(30) FIGS. 30 and 31 are schematic front elevation views of an embodiment of an apparatus for use in forming a subcutaneous anatomical tunnel for implantation of a loop vascular graft using two-piece tunneling device having a distal locking mechanism.

(31) FIG. 32 is a side perspective view of an embodiment of an apparatus for use in forming a subcutaneous anatomical tunnel for implantation of a vascular graft using a helical tunneling device.

(32) FIG. 33 is a side perspective view of an embodiment of a tunnel forming apparatus showing a guide wire and a sheath.

(33) FIG. 34 is a side perspective view of the tunnel forming apparatus as shown in FIG. 30 with the guide wire fully received within the sheath.

(34) FIG. 35 is a transverse cross-section view of the tunnel forming apparatus as shown in FIG. 31
(35) FIG. 36 is a side perspective view of the tunnel forming apparatus as shown in FIG. 31 with the sheath inflated and a vascular graft fully received within the sheath.
(36) FIG. 37 is a transverse cross-section view of the tunnel forming apparatus as shown in FIG. 33.

DESCRIPTION

(37) Certain terminology is used herein for convenience only and is not to be taken as a limiting. For example, words such as “upper,” “lower,” “left,” “right,” “horizontal,” “vertical,” “upward,” “downward,” “top” and “bottom” merely describe the configurations shown in the FIGs. Indeed, the components may be oriented in any direction and the terminology, therefore, should be understood as encompassing such variations unless specified otherwise. The words “interior” and “exterior” refer to directions toward and away from, respectively, the geometric center of the core and designated parts thereof. The terminology includes the words specifically mentioned above, derivatives thereof and words of similar import.

(38) Referring now to the drawings, wherein like reference numerals designate corresponding or similar elements throughout the several views, an embodiment of an apparatus for use in forming a subcutaneous anatomical tunnel for implantation of a vascular graft in a patient is shown in FIGS. 1-3. The tunnel forming apparatus, also referred to herein as a tunneling device, comprises a tunneling instrument 100 including a proximal handle 102 and a distal tip assembly 50.

(39) An arteriovenous graft suitable for use in this application is described in commonly owned U.S. Pat. No. 9,585,998, the contents of which are hereby incorporated by reference in their entirety. It is understood that the tunnel forming apparatus is also capable of use with other vascular grafts as well as a natural tissue graft or fistula.

(40) An embodiment of the tip assembly 50 is shown in FIGS. 4 and 5. The tip assembly 50 comprises a tip 51 and a swivel connector 58. The tip 51 and the swivel connector 58 may be constructed of stainless steel, but one of ordinary skill in the art will recognize that other materials may be suitable to connect an arteriovenous graft to the tunneling device. One such example of an alternative material is a plastic, such as a hard plastic.

(41) Referring to FIGS. 6 and 7, the tip 51 includes a bullet shaped distal end 52 and a longitudinally extending proximal end coupler 54. The swivel connector 58 comprises a bell-shaped ferrule 59 defining a central axial channel 63 for rotatably receiving the coupler 54 (FIGS. 8 and 9). The proximal end of the coupler 54 defines an internally threaded axial opening so that the ferrule 59 may be secured to the coupler 54 by a screw 61 for an axially rotatable swivel connection.

(42) The proximal end of the ferrule 59 is externally threaded 65 for securing the tip assembly 50, and associated vascular graft, to the internally threaded distal end of the tunneling instrument 100. When assembled, the swivel connector 58 is rotatable about its longitudinal axis on the coupler 54 allowing the tip assembly 50 to be threaded into the tunneling instrument 100 without rotating a connected graft or a graft covered by a sheath. Other conventional methods of fastening the swivel connector 58 to the tunneling instrument 100 include snap-on or clip-on techniques that allow the ferrule 59 to snap or clip into a tunneling instrument. These and other fastening techniques are contemplated to be within the scope of the invention.

(43) A fastening clip may be used to couple a vascular graft 10, or the graft and a sheath 20, to the tip assembly 50 such that the sheath is attached to and encloses at least the proximal end and a portion of the length of the vascular graft. In one embodiment shown in FIG. 10 and generally designated at 70, the fastening clip comprises a hollow cylindrical cap 70. A distal end of the cap 70 includes a plurality of fingers 72 extending radially inwardly into the passage 71 defined by the cap. The cap 70 is configured to receive and enclose a portion of the proximal end of the arteriovenous graft. Referring to FIGS. 11 and 12, the tip 51 includes a smaller diameter portion spaced from the distal bullet-shaped end 52 and forming a peripheral annular groove 56. The

annular groove **56** is configured to receive the proximal end of the graft, and sheath if present, enclosed by the cap **70**. In use, the graft and sheath may be coupled to the tip **51** by passing the graft and sheath through the cap **70**. The graft and sheath are then secured to the tip **51** by slipping the cap **70** with the graft and sheath over the distal end **52** of the tip **51** and axially along the tip **51** until the fingers **72** seat in the groove **56**. In this configuration, the proximal end of the graft and sheath are enclosed within the cap **70** and securely within the groove **56** for fixedly positioning the graft and sheath therein.

(44) Another embodiment of a fastening element for securing an arteriovenous graft, or the vascular graft and a sheath, to the tip assembly **50** is shown in FIGS. **13-16** and generally designated at **80**. In this embodiment, the fastening element comprises a lock ring **80** mounted around a proximal end of the graft, or graft and sheath, for securing the graft in the groove **56** and preventing axial movement of the graft relative to the tip assembly **50**. More particularly, the lock ring **80** may be coupled to the tip **51** by crimping about a pivot pin **81** such that teeth **82** on opposed surfaces of the free ends of the lock ring **80** engage for enclosing the graft and sheath within the groove **56** and fixedly positioning the graft, or graft and sheath therein (FIGS. **17** and **18**). This arrangement provides for coupling of a tunneling instrument **100** to an arteriovenous graft **10** by a compressive force on the graft for subcutaneously deploying the graft **10** in a patient. The arteriovenous graft **10** may be surrounded by a sheath **20** for drawing both into the subcutaneous tissue cavity in the patient. The sheath **20** is secured to the tip assembly **50** and surrounds at least a substantial portion of the arteriovenous graft as shown in FIGS. **21** and **22**.

(45) FIG. **19** shows yet another embodiment of a fastening element for securing an arteriovenous graft, or the graft and a sheath, to a tunneling instrument **100** and is generally designated at **120**. In this embodiment, the tip assembly **50** is removed and the fastening element is secured directly to the distal end of the tunneling instrument **100**. The fastening element comprises a coupler **120** including a socket **122** and a clip **124** to be connected between a distal end of the tunneling device and a proximal end of the graft. The socket **122** includes a hollow cylindrical sleeve **126**, a washer **128** and a threaded proximal plug **130** for threadably securing the sleeve **126** to an internally threaded end of the tunneling device **100**. The plug **130** is connected to the proximal end of the sleeve **126** via an opening **127** in the sleeve with a screw **131** and washer **128** there between. This arrangement allows the plug **130** to rotate relative to the sleeve **126**. The sleeve **126** has opposed partially circumferential slots **132** spaced intermediately along the length of the sleeve **126**. The proximal end of the clip **124** includes a pair of axial flexible arms **134**. Each of the arms **134** has a radially outwardly extending shoulder **136**. The remainder of the clip **124** distally of the arms **134** is generally circular in cross-section.

(46) In use, the plug **130** and connected sleeve of the socket **122** is threaded onto the distal end of the tunneling instrument **100**. The proximal end of the graft is pushed over the distal portion of the clip **124** for securing the graft to the clip. The clip **124** and graft are then pushed into the distal open end of the sleeve **126**. This causes the arms **134** to compress inwardly together until the shoulders **136** reach the slots **132** at which point the arms **134** spring outwardly as the shoulders **136** enter the slots **132**. This fastens the socket **122** and clip **124** together along with the connected tunneling device and graft. The user can now pull the graft into the subcutaneous tunnel previously formed by the tunneling instrument **100**.

(47) Referring now to FIG. **20**, another embodiment of a fastening element for securing an arteriovenous graft, or the graft and a sheath, directly to a tunneling instrument **100** is shown and generally designated at **90**. The fastening element comprises a frustoconical cup **90** has an axial opening through a small diameter externally threaded proximal end **92** configured for receiving the graft (Step A). The proximal end of the graft passes through the end **92** of the cup **90** (Step B) and is then folded back over the threads (Step C). The proximal end **92** of the cup **90** is threaded into the distal end of the tunneling instrument **100** (Step D), such that the large diameter open end of the cup **90** is facing distally away from the tunneling device **100** (Step E).

(48) In use, after the distal end of the tunneling instrument **100** emerges from the second distal incision, the proximal end of the vascular graft is pulled through the cup **90** and everted back over the periphery of the threaded end **92** of the cup **90** (Steps B and C). The cup **90** and graft are then threadably connected with the distal end of the tunneling device **100** for securing the end of the graft (Steps D and E). Once connected, the cup **90** is pulled by the tunneling device **100** along with the graft through the anatomical subcutaneous path created by the tunneling device. The cup **90** assists in the passage of the graft into the body by causing the outward deflection of surrounding tissue upon contacting the frustoconical cup **90**. After the graft is pulled through the tissue tunnel, the cup is removed prior to anastomosis of the graft. It is understood that other means for connecting the end of the cone **90** to the tunneling device **92** which do not require threads are suitable.

(49) Referring to FIGS. 23A-23C, a system is shown for implanting an arteriovenous graft (AVG) **10** further comprising a tubular sheath **20** positioned over and enclosing at least a substantial portion of the outer surface of the graft during the implantation procedure. Following implantation of the arteriovenous graft **10**, the sheath **20** is configured to roll along the graft **10** upon application of a pulling force along the longitudinal axis of the sheath **10** to extract the sheath.

(50) The sheath **20** comprises a tubular structure that is dimensioned in diameter and length to cover at least a substantial portion of the outer surface of the AVG **10**. The sheath **20** can be constructed from any smooth, flexible and compressive biocompatible material. Suitable material can be porous, non-porous, permeable, or impermeable. Examples of such materials include, but are not limited to, silk, silicone, fluoropolymers such as expanded polytetrafluoroethylene (ePTFE), high density polyethylene (HDPE), and other polymers such as polyesters and polyimides. Various desired configurations may be achieved by varying film materials and characteristics, such as thickness and width. The sheath **20** may be extruded, for example, directly over a lead body or as a pre-manufactured item subsequently attached to a lead. The sheath **20** allows for easier insertion of the graft **10** through the tunneled tissue path due to the unrestricted and flexible nature of the compressible outer surface of the sheath. The sheath **20** may be constructed of a material that does not excessively flex, so that the sheath **20** absorbs the tensile force imparted during tunneling as opposed to the graft **10**. This will help to prevent damage to the graft **10**, which may reduce graft stretching and subsequent graft weeping and seroma formation, and damage that may allow the graft to kink.

(51) In the embodiment of the implantation system shown in FIGS. 23A-23C, the sheath **20** is doubled over on itself such that the amount of material used to make the sheath will be at least double the length desired to cover the AVG. The sheath **20** in this embodiment has a “double walled” construction. An inner portion **22** of the sheath **20** extends from a position proximal of the AVG **10** radially over and axially along the AVG to a position distal of the AVG where the inner portion **22** is folded back to provide an outer portion **24**. The outer portion **24** extends radially over the inner portion **22** and to a position proximal of the AVG **10**. When the sheath **20** is extracted by a rolling action, the sheath **20** will slide over itself thus reducing the force necessary to extract the sheath. It is understood, however, that an embodiment of the sheath **20** may have a sufficiently low coefficient of friction that the sheath can be removed from the tunnel by pulling the sheath from the tunnel and not a rolling motion.

(52) As shown in FIGS. 23A-23C, the sheath **20** has perforations **30** disposed along the longitudinal axis of the sheath. The perforations **30** are generally oriented linearly in diametrically opposed positions along the length of the sheath **20**. The perforations **30** allow the sheath to tear at the perforations upon rolling as the sheath **20** is extracted. In another embodiment, the sheath **20** may comprise weakened areas of material that will tear as the sheath is extracted. During retraction, the sheath **20** is drawn axially proximally relative to the AVG **10** to retract the sheath from over the AVG. The pulling force retracts the sheath **20** from the distal end and progressively proximally with the outer sheath **24** moving over the inner sheath **22** in a rolling manner.

(53) Referring to the embodiment shown in FIGS. 24A and 24B, the sheath **20** is drawn axially proximally relative to the AVG **10** to retract the sheath from over the arteriovenous graft by the pulling force provided by a moveable tether **40**. The tether **40** extends axially from adjacent a proximal end of the arteriovenous graft to an opposite distal end of the graft where the tether **40** joins the outer portion **24** of the sheath **20** at or beyond the distal end of the graft **10**. The tether **40** extends sufficiently proximally externally to the first incision so that it can be pulled upon for retracting the sheath **20**. The tether **40** is pulled upon to retract the sheath **20** from the distal end and progressively proximally with the outer sheath **24** moving over the inner sheath **22** in a rolling manner. The tether **40** may be a plastic thread or a metal wire.

(54) Referring to the embodiment shown in FIGS. 25A-25C, the sheath **20** is a single layer having perforations **30** disposed along the longitudinal axis of the sheath. The perforations **30** are generally oriented linearly in close parallel relation along the length of the sheath **20**. The perforations **30** allow the sheath to tear at the perforations upon rolling as the sheath **20** is extracted. The tether **40** extends axially from adjacent a proximal end of the arteriovenous graft to an opposite distal end of the graft where the tether **40** joins the sheath **20** between the perforations **30** at or beyond the distal end of the graft **10**. The tether **40** extends sufficiently proximally externally to the first incision so that it can be pulled upon for retracting the sheath **20**. The tether **40** is pulled upon to retract the strip of the sheath **20** between the perforations **30**.

(55) When the arteriovenous is subcutaneously located as desired, the tether **40** or the sheath **20** itself is pulled. FIGS. 23B, 24B and 25B show the pulling force as depicted by arrows. The pulling force moves the sheath **20** proximally which, in one embodiment, causes the outer sheath **24** to slide over the inner sheath **22**. As the sheath **20** rolls proximally, the sheath is torn at the perforations **30**. As the pulling force continues, the sheath **20** will continue to move proximally until the sheath **20** is extracted from the first incision and the arteriovenous graft **10** is uncovered.

(56) The sheath **20** may be coated on the outside surface with a lubricious substance to provide a low coefficient of friction, aiding movement of the sheath **20** and graft **10** through tissue when pulled by a tunneling instrument. This will minimize tissue drag and tissue trauma during insertion of the arteriovenous graft **10** or removal of the sheath **20** after implantation. The lubricant will also allow the sheath **20** to slide smoothly across itself. Solid lubricants (i.e. graphite, waxes, silicone), fluid lubricants (i.e. hydrocarbon oils, silicone oils), gels (i.e. hydrogel) or any other biocompatible material known in the art may be used. In one embodiment, the sheath **20** can be coated or wetted immediately before implantation by the user. In another embodiment, the invention comprises a kit comprising a sheath and a wetting agent for wetting the sheath **20**. In another embodiment, the invention comprises a kit comprising a sheath **20**, an arteriovenous graft **10** and a wetting agent for wetting the sheath.

(57) The sheath **20** may be attached to at least the distal end of the arteriovenous graft by mechanical, rail or interference fit, mechanical structures, heat bonding or by a biocompatible adhesive or other securing means. Example adhesives are thermoplastic fluoropolymers, such as fluorinated ethylene propylene (FEP). The sheath may also be attached to the arteriovenous graft following its manufacture as a separate component.

(58) The system and method for implanting the arteriovenous graft, including a sheath enclosing the graft during implantation, have many advantages, including atraumatic implantation of an arteriovenous graft and subsequent extraction of the associated sheath. The sheath provides a flexible, compressible outer surface for the graft that may allow for easier insertion of the graft into the tissue cavity with less trauma, less friction, and less drag during placement. Thus, the system and method described herein reduce damaging forces to surrounding tissues associated with the implant procedure and minimize the resultant trauma to this tissue and its healing response. Due to the smoothness and collapsible low profile of the sheath, the tunneling procedure may be faster and easier to use, in addition to being less traumatic to tissue.

(59) FIG. 26 shows an embodiment of a tunneling instrument generally designated at **150**. The

shaft **152** defines an axial opening extending through its entire length. The distal end of the shaft **152** has a plurality of holes **156** in fluid communication with the axial opening through the shaft **152**. This configuration allows infusion or aspiration of a fluid from a fluid source delivered through the shaft **152** to the tip **154**. During advancement of the tunneling instrument **150** through the subcutaneous tissue, fluid can be infused from the fluid source and through the distal shaft holes **156**. The infused fluid exits the shaft holes **156** adjacent to the distal end **153** of the tip **154** for lubricating the tissue passage. Examples of fluid for use in this application include, but are not limited, to phosphate-buffered saline, saline and buffered saline as well as gas.

(60) Referring to FIG. **27**, another embodiment of a tunneling instrument for delivering fluid is shown and generally designated at **160**. In this embodiment, the tip **162** or nose section has a plurality of openings **164**. When the tip **162** is fastened to the distal end of the shaft **152**, the axial opening through the shaft is in fluid communication with the holes **164** in the tip **162**. In use, the tunneling apparatus **160** delivers pressurized fluid, such as air, from a fluid source through the shaft **152** and through the holes **164** in the tip **162** during advancement of the tunneling apparatus **160** through subcutaneous tissue. The pressurized fluid exits the tip **162** in the direction of advancement of the tunneling apparatus **160** for aid in dissecting tissue during the tunneling procedure.

(61) Yet another embodiment of a tunneling apparatus is shown in FIG. **28** and generally designated at **170**. The tunneling apparatus **170** comprises a rigid shaft **172** with a magnetic tip **174**, or nose section, at a distal end **173** of the shaft. As with other tips described herein, the magnetic tip **174** has an elliptical or circular shape that facilitates the blunt dissection inherent in the tunneling procedure. As shown in FIG. **28**, a magnetic wand **176** is provided for engaging the tip **174** across the skin **158** boundary. During the tunneling procedure, the tip **174** is advanced through the subcutaneous tissue by moving the wand **176** along the surface of the skin **178**.

(62) An embodiment of a self-guiding tunneling apparatus is shown in FIG. **29**. In this embodiment, markers **92** placed on the skin **94** of a patient indicate a predetermined path for the anatomical tunnel for placement of a vascular graft **10**. The markers **92** identify proper placement of a tunneling device (not shown) beginning from an incision **96** into the body. The user follows the markers **92** in navigating the tunneling device through the body, thereby increasing the accuracy of the procedure by following well-established practices of guide wire navigation.

(63) An embodiment of a locking mechanism for use with a looped vascular graft is shown in FIGS. **30** and **31** and generally designated at **800**. The locking mechanism **800** is for use with a pair of arcuate tunneling rods **802** schematically shown in FIG. **30**. The rods **802** are arranged to be slid along curved paths that mirror one another. The length of the rods **802** depends on the procedure, and particularly on the length of the vascular graft to be inserted. The distal end of each rod **802** is inserted into the exposed tissue at an incision **804**. The rods **802** are advanced to a point spaced from the incision **804** with the proximal ends **803** of the rods **802** remaining outside of the body. The distal ends **805** of the rods **802** each carry a portion of the locking mechanism **800**. In assembling the pair of rods **802**, the locking mechanism **800** components are joined subcutaneously where the distal ends **805** of the rods **802** come together.

(64) A further embodiment of a tunneling apparatus is shown in FIG. **32** and generally designated at **180**. The tunneling apparatus **180** comprises a rigid shaft **182** with a tip **184**, or nose section, at a distal end **183** of the shaft. In this embodiment, the shaft **182** is fabricated in a helical shape. Thus, the anatomical tunnel formed through the tissue follows a helical path. The vascular graft is then placed within the tissue tunnel in the form of a helical spring.

(65) Yet another embodiment of a tunneling apparatus is shown in FIGS. **33-37** and generally designated at **400**. The tunneling apparatus **400** comprises a guide wire **402** and an elongated tubular sheath or conduit **404**. The guide wire **402** is substantially rigid for use as a tunneling instrument in creating a subcutaneous tissue path in the patient. The diameter of the guide wire **402** is smaller in comparison to a conventional tunneling instrument to reduce the trauma to the surrounding tissue as the guidewire is pushed through the tissue. The sheath **404** is mounted over

the guide wire **402** and encloses the guide wire during a tunneling procedure into the tissue of a patient.

(66) The sheath **404** is formed from a double layer of thin flexible, compressible material closed at both ends and is radially expandable. Accordingly, the sheath **404** has a contracted deployment state, in which the sheath is compressed against the guide wire **402** for use as an insertion tool assembly, and a radially expanded inflated state, which the sheath **404** assumes upon being inflated with a fluid, including a liquid or a gas. The guide wire **402** functions as a means for supporting the sheath **404** in its contracted state during the tunneling step when the tissue is traversed. After the creating the tunnel in tissue, the sheath **404** is inflated, expanding radially outwardly pushing against the tissue walls and providing a clear central hole. In the inflated condition, the sheath **404** is configured for providing an opening of a diameter through which a distal end **403** of the guide wire **402** may be withdrawn for receiving a vascular graft **10** with the application of a reasonable level of axial force. The application of radial force as opposed to longitudinal force in forming the area of the subcutaneous tunnel minimizes trauma to the surrounding tissue and therefore reduces the rate of swelling, pain, and bleeding post-procedure. This facilitates immediate or early cannulation of an implanted vascular graft. Additionally, minimizing the force on the graft **10** during implantation will help to prevent damage to the graft, which may reduce graft stretching and subsequent graft weeping and seroma formation, and help prevent graft kinking.

(67) The sheath **404** may be fabricated from polyethylene, polyvinyl chloride, or other suitable materials well known to those skilled in the art. These materials may further include, but are not limited to, Mylar ribbon, Teflon ribbon, and polypropylene. The smooth outer surface of the material of the sheath **404** may allow the tunneling device **400** to have a low coefficient of friction. The result is easier insertion of the guide wire **402** through the tissue with less trauma, less friction, less blunt dissection and less drag during placement. Due to the smoothness and collapsible low profile of the sheath **404**, the sheath does not substantially increase the outside diameter of the tunneling apparatus **400** for deployment. The tunneling apparatus **400** may thus be faster and easier to use, in addition to being less traumatic to tissue. The sheath **404** may also be coated with a lubricous material to further decrease its coefficient of friction and facilitate insertion of the tunneling device **400**. The coating aids movement of the sheath **404** and guide wire **402** through the tissue while minimizing tissue drag and trauma during insertion or during sheath **404** removal after implantation. It is understood that a wide variety of coatings are available, including therapeutic agents for delivery of therapeutic materials.

(68) In use, two spaced incisions are first made through the patient's skin into underlying tissue. The distal end **403** of the guide wire **402** with the sheath in the deployment state is inserted into a first incision and then forced horizontally through the subcutaneous tissue along a path until the distal end **403** exits the second incision. The distal end **403** of the guide wire **402** is then attached to a proximal end of the vascular graft **10** by the surgeon, such as, for example, using simple mechanical means, a compression fit collar, staples or sutures or other fastening techniques acceptable for implantation within the tissue of the body. After attachment of the graft **10** to the guide wire **402**, the sheath **404** is inflated with fluid forcing the sheath against the dissected tissue defining the tunnel. The expansion of the sheath **404** forms an internal passage through the sheath that is sufficiently large to allow the subsequent withdrawal of the guide wire **402**. The guide wire **402** is retracted proximally, which simultaneously draws the attached graft **10** into and through the lumen of the inflated hollow sheath **20** until the proximal end of the graft **10** exits the first incision. The inflated sheath **204** allows the surgeon to easily pull the vascular graft **10** through the internal lumen defined by the sheath **204**, which is substantially oversized in comparison to the outside diameter of the vascular graft **10**. Once the graft **10** has been drawn to the site of the first incision, the proximal end of the graft **10** is disconnected from the distal end **203** of the guide wire **202**.

(69) Next, the sheath **204** is deflated and extracted from the subcutaneous tissue path without extracting the graft **10**. Specifically, the surgeon holds the graft **10** in place by grasping a distal end

of the graft extending from the second incision and pulls the sheath **204** through the first incision. The graft **10** remains within the anatomic subcutaneous tunnel. With the sheath **204** removed, the tissue collapses against the outer surface of the graft **10**. The surgeon now forms anastomoses at each end of the graft **10** by suturing the ends of the graft to a blood vessel at the desired locations.

(70) The embodiments of the tunneling apparatus as described herein are shown in use for procedures with a vascular graft suitable for implantation in the body and used to reestablish or redirect the flow of blood beyond the blockage area. It is understood that the several tunneling apparatus described and shown herein may be used in other surgical implantation procedures requiring placement of the vascular graft within the subcutaneous tissue. One of ordinary skill in the art will recognize that the embodiments of the tunneling apparatus as described are not directed to a specific vascular graft design, but are generically applicable to many different types of vascular grafts, which may be a synthetic graft constructed from different materials or a natural tissue graft.

(71) The tunneling apparatus and methods as described have many advantages, including a reduction of friction and resistance, and related tissue damage, during the passage of the tunneling device when forming an anatomical tunnel. The tunneling apparatus and methods may help to reduce swelling and damage to the surrounding tissue, which would facilitate immediate or early cannulation of a subsequently implanted vascular graft.

Claims

1. An implantation apparatus for subcutaneous implantation in a patient using a tunneling instrument of the type including a shaft having a proximal end and a distal end and a tip assembly positioned at the distal end of the tunneling instrument, the implantation apparatus comprising: a vascular arteriovenous graft having a proximal end and a distal end and a length between the proximal end and the distal end; and a coupler adapted to couple the distal end of the tunneling instrument and the proximal end of the graft, the coupler comprising a plug configured to be coupled to the distal end of the tunneling instrument, a clip configured to be coupled to the proximal end of the graft, and a socket for a rotatable connection of the clip to the plug such that the plug is rotatable about its longitudinal axis relative to the clip to facilitate attachment of the graft to the tunneling instrument, wherein the coupler is configured for connecting the graft to the tunneling instrument upon removal of the tip assembly.
2. The implantation apparatus as recited in claim 1, wherein the coupler comprises a screw thread formed on an external surface for connection of the coupler to the tunneling instrument.
3. The implantation apparatus as recited in claim 1, wherein the coupler comprises a ferrule defining an axial passage for rotatably receiving a proximal end of the tip.
4. The implantation apparatus as recited in claim 1, further comprising a removable sheath covering the graft.
5. The implantation apparatus as recited in claim 4, wherein the sheath is configured to substantially cover the length of the graft.
6. The implantation apparatus as recited in claim 1, wherein the tip comprises a body including an intermediate portion having a reduced diameter forming a circumferential groove around a periphery, and wherein the clip is positioned in the groove, a portion of the graft being positioned between the clip and the tip within the groove, the clip being crimped about the tip to fix the graft to the tip.
7. The implantation apparatus as recited in claim 6, wherein the clip is sized and shaped to provide a snap-fit connection within the groove.
8. The implantation apparatus as recited in claim 1, wherein the tip is adapted to be removably attached to the distal end of the tunneling instrument.
9. The implantation apparatus as recited in claim 8, wherein the coupler comprises a frustoconical cone having an axial opening for receiving the graft, a length of the cone including a screw thread

formed on an external surface for connection of the cone to the tunneling instrument, wherein a portion of the graft is positioned between the threads of the cone and the tunneling instrument to fix the graft to the tip.

10. The implantation apparatus as recited in claim 8, wherein the coupler comprises a hollow sleeve for rotatable connection to the tunneling instrument, and a plug received within the proximal end of the graft, the sleeve configured for receiving the plug such that a portion of the graft is positioned between the plug and the sleeve to fix the graft in the sleeve.

11. The implantation apparatus as recited in claim 1, wherein the graft comprises a synthetic vascular graft.

12. The implantation apparatus as recited in claim 1, wherein the graft comprises a natural tissue vascular graft.

13. An implantable device for subcutaneous delivery in a patient using a tunneling instrument of the type including a shaft having a proximal end and a distal end, the implantable device comprising: a vascular graft having a length, a distal end, a proximal end, an outer surface and a longitudinal axis; and a sheath having a length and being positioned over a substantial portion of the outer surface of the vascular graft, the sheath including a tubular member, wherein the tubular member includes an inner portion and an outer portion, the outer portion being an everted portion of the tubular member, the sheath having longitudinally spaced perforations along the length of the sheath for permitting tearing of the sheath; and a tether coupled to the sheath, wherein the sheath is configured to evert upon application of a longitudinal force to the tether following implantation of the vascular graft to cause the sheath to move in a proximal direction during removal of the sheath from the implanted graft, and wherein the sheath has a material strength such that the sheath everts upon application of the longitudinal force without tearing the sheath and upon eversion and continued actuation of the tether that applies a force to the sheath exceeding a tearing strength of the sheath the sheath tears progressively along the perforations that are generally oriented in a linear arrangement along the length of the sheath.

14. The implantable device as recited in claim 13, wherein two rows of perforations are generally oriented circumferentially opposite to one another.

15. The implantable device as recited in claim 13, wherein two rows of perforations are generally oriented adjacent and parallel to one another.

16. The implantable device as recited in claim 13, further comprising a tether connected to the sheath for applying a pulling force in a proximal direction.

17. The implantable device as recited in claim 16, wherein the one end of the tether is attached to the sheath at a position adjacent to the distal end of the sheath.

18. An implantable device for subcutaneous delivery in a patient using a tunneling instrument including a shaft having a proximal end and a distal end, the implantable device comprising: a vascular graft having a length, a distal end, a proximal end, an outer surface and a longitudinal axis; and a sheath having a length and being positioned over a substantial portion of the outer surface of the vascular graft, the sheath including an inner portion and an outer portion, the inner portion configured to evert to transition to the outer portion, the sheath having longitudinally spaced perforations along the length of the sheath for permitting tearing of the sheath, wherein the sheath is folded back on itself to provide a double wall including an inner portion and an outer portion, and wherein the sheath is configured to slide over itself upon application of a longitudinal force to the sheath following implantation of the vascular graft and wherein the sheath has a material strength such that the sheath everts upon application of the longitudinal force without tearing the sheath and after the sheath is everted continued actuation of the sheath is configured to cause the sheath to tear along the perforations and move in a proximal direction during removal of the sheath from the implanted graft thus reducing the force necessary to extract the sheath.
